

Model Rocketry 2026-2027

Critical Design Review (CDR)

Team ID: 2026-INSPACe-ROCKETRY-059
SRMIST, TRICHY

Propulsion Subsystem Design — Motor Selection & Technical Summary

Parameter	Motor A — TIVRA K-925
Total Impulse	2500 Ns
Average Thrust	704 N
Peak Thrust	1559 N
Burn Time	3.6 s
Motor Weight	4459 g
Vehicle Liftoff Weight	15,000 g

147.15 N
15KG WET WEIGHT

4.8
AVG T/W RATIO

10.6
PEAK T/W RATIO

1. Motor Selection & Class

- Selected Configuration:** TIVRA K-925 (Part No. K-925, Class K Impulse) chosen for Aluminum body class.
- Mass Ceiling:** Aligned with the 15,000 g vehicle liftoff mass cap.
- Competition Compliance:** 2500 Ns impulse strictly satisfies the <2800 Ns cap (Req. 1.7).
- Alternative Status:** Motor B (K-560 / 11.5 kg PVC class limit) is officially rejected.

3. Thrust Generation Mechanism

- 1. Combustion & Gas Generation**
Propellant ignition produces high-temperature, high-pressure gases.
- 2. Nozzle Supersonic Expansion**
Gases accelerate through converging-diverging nozzle.
- 3. Force Transfer & Flight Profile**
Reaction force transfers to airframe via motor mount (3.6s burn into coast).

2. Mass Budget Reconciliation

- PDR Baseline Update:** Replaces preliminary 6.6 kg liftoff mass baseline.
- Actual Vehicle Modeling:** Simulations must use finalized K-925 configuration mass rather than assumes 15 kg limit.
- Subsystem Updates:** Re-calculate CG/CP stability, apogee/altitude prediction, and structural loads.
- Team Action:** Rocket Architecture & Recovery sub-teams to finalize mass reconciliation before CDR.

4. Data Handoff & Interfaces

- Architecture / Aerodynamics:** Provide verified K-925 motor mass (4459g) & CG envelope for structural checks.
- Recovery Subsystem:** Compute post-burnout dry mass using verified K-925 propellant burn data.
- Forward Interface:** Retention structures must comply with K-925 mechanical specs in CAD & BOM.



VEHICLE GENERAL ASSEMBLY

SYSTEM SPECIFICATIONS

Airframe Outer Diameter (OD): Ø100.0 mm

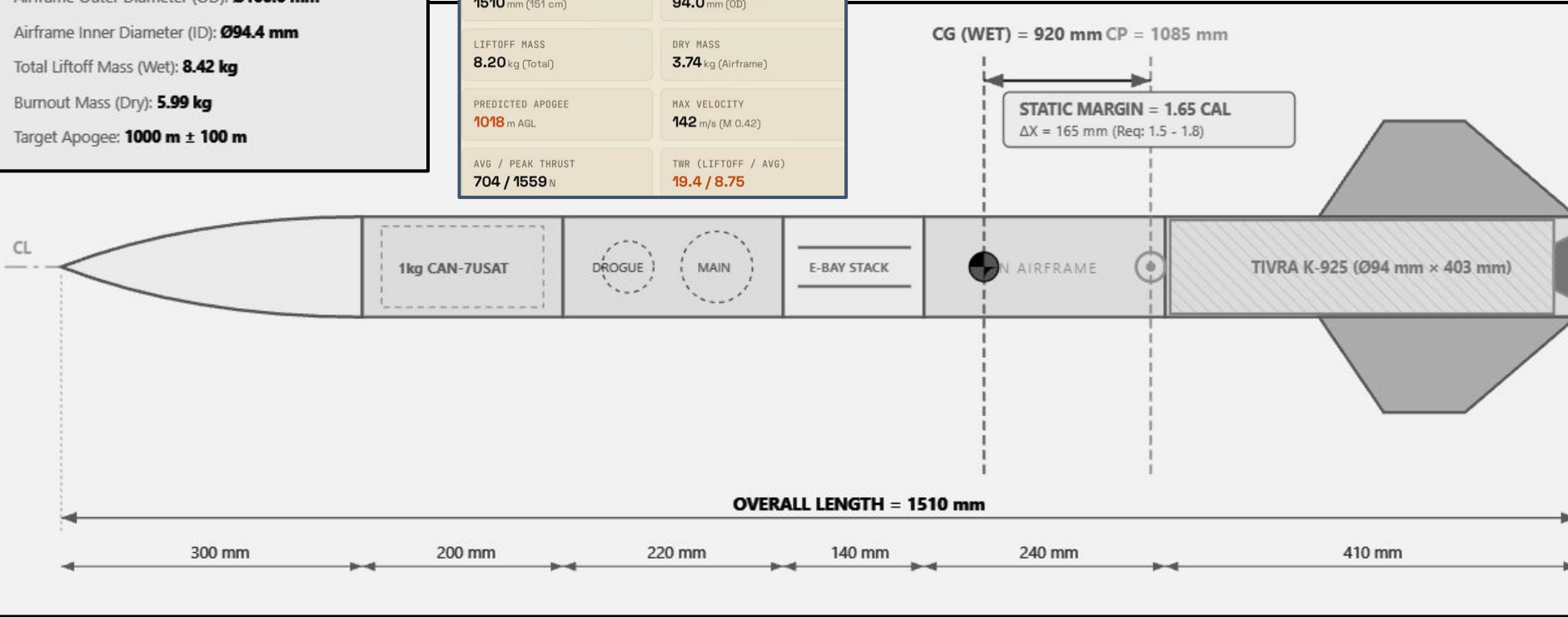
Airframe Inner Diameter (ID): Ø94.4 mm

Total Liftoff Mass (Wet): 8.42 kg

Burnout Mass (Dry): 5.99 kg

Target Apogee: 1000 m ± 100 m

OVERALL LENGTH 1510 mm (151 cm)	AIRFRAME CALIBER 94.0 mm (OD)
LIFTOFF MASS 8.20 kg (Total)	DRY MASS 3.74 kg (Airframe)
PREDICTED APOGEE 1018 m AGL	MAX VELOCITY 142 m/s (M 0.42)
AVG / PEAK THRUST 704 / 1559 N	TWR (LIFTOFF / AVG) 19.4 / 8.75

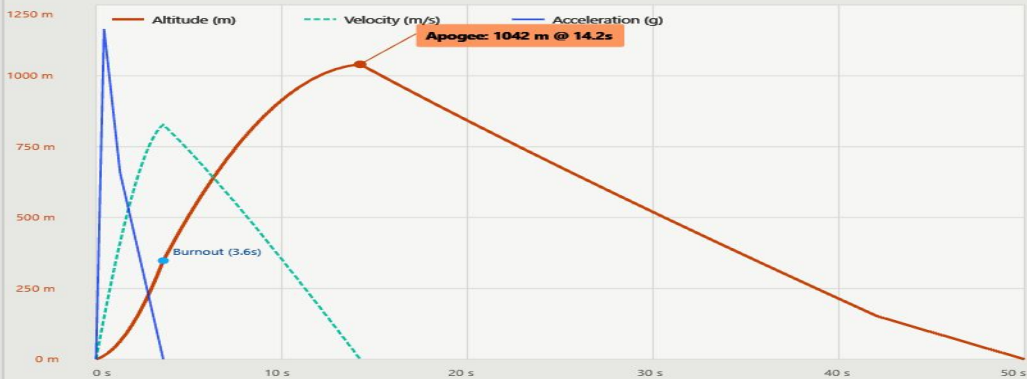




DESIGN CALCULATION & STABILITY



Flight Simulation Trajectory (TIVRA K-925 Motor)



Simulation Summary & Design Criteria

Target Apogee:	1042 m (Bracket: 1000 ± 100m)
Static Margin (Liftoff):	1.65 Cal (Req: 1.5 - 1.8 Cal)
Static Margin (Burnout):	2.75 Cal
Max Velocity / Mach:	182.5 m/s (Mach 0.54 @ Burnout)
Max Acceleration:	16.8 g (164.8 m/s²)
Launch Guide Clear Vel:	26.4 m/s (Rail length: 3.0 m)
Main Deployment / Landing:	150 m AGL / 5.4 m/s Touchdown

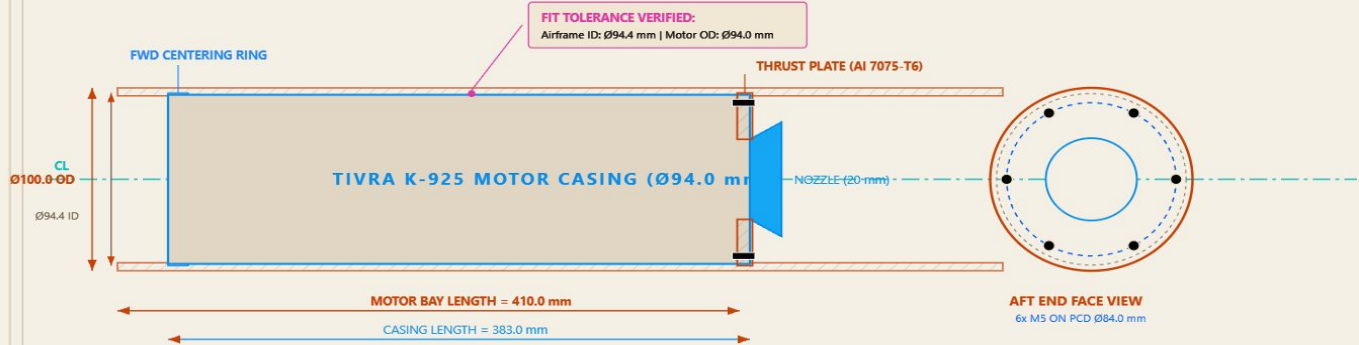
MASS & STABILITY BREAKDOWN		Total: 8420 g
Structure (Dry Airframe + Fins):	2180 g	
CAN-7USAT Scientific Payload:	1000 g	
Avionics Stack & Recovery Rig:	781 g	
TIVRA K-925 Motor (Casing + Prop):	4459 g	



MOTOR BAY INTERFACE & THRUST STRUCTURE ASSEMBLY

RECONCILED DUAL-CLEARANCE TIVRA K-925 INTEGRATION • SECTION A-A

DOC NO: AF-008-REV-C



BILL OF MATERIALS (BOM) - MOTOR BAY

- ITEM 1: Main Airframe Tube (Fiberglass / Epoxy) - Qty 1
- ITEM 2: Forward Centering Ring (Phenolic Canvas) - Qty 1
- ITEM 3: Aft Thrust Retainer Ring (Aluminium 7075-T6) - Qty 1
- ITEM 4: Socket Head Cap Screws M5×16 (SS304 A2-70) - Qty 6
- ITEM 5: Threaded Brass Inserts M5×10 (Bonded) - Qty 6
- ITEM 6: High-Temp Structural Epoxy (Araldite 2014-1)

MANUFACTURING TOLERANCES

- Airframe Inner Diameter (Bore): $+0.20 / -0.00$ mm
- Airframe Outer Diameter: ± 0.15 mm
- Centering Ring Outer Diameter: ± 0.05 mm
- Bolt Hole PCD Diameter: ± 0.10 mm
- End Face Squareness to Bore Axis: ≤ 0.15 mm TIR
- Thrust Face Runout (Peak Load): ≤ 0.05 mm

STRUCTURAL LOAD QUALIFICATION

- Peak Motor Thrust (\$F_{pk}\$): **1559 N**
- Average Cruise Thrust (\$F_{avg}\$): **704 N**
- Thrust Plate Compressive Stress: 12.4 MPa (Yield: 503 MPa)
- M5 Fastener Shear Safety Factor (\$SF_S\$): **4.8** (> 2.0 Req)
- Airframe Column Buckling Margin: **> 12 kN Critical Load**

VEHICLE SPECIFICATIONS

OVERALL LENGTH

1510 mm (151 cm)

LIFTOFF MASS

8.20 kg (Total)

PREDICTED APOGEE

1018 m AGL

AVG / PEAK THRUST

704 / 1559 N

AIRFRAME CALIBER

94.0 mm (OD)

DRY MASS

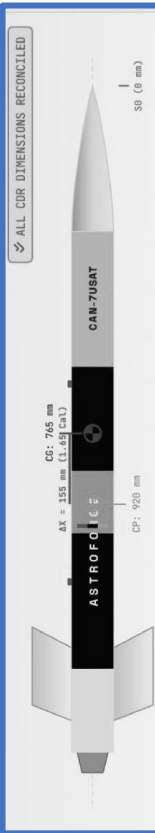
3.74 kg (Airframe)

MAX VELOCITY

142 m/s (M 0.42)

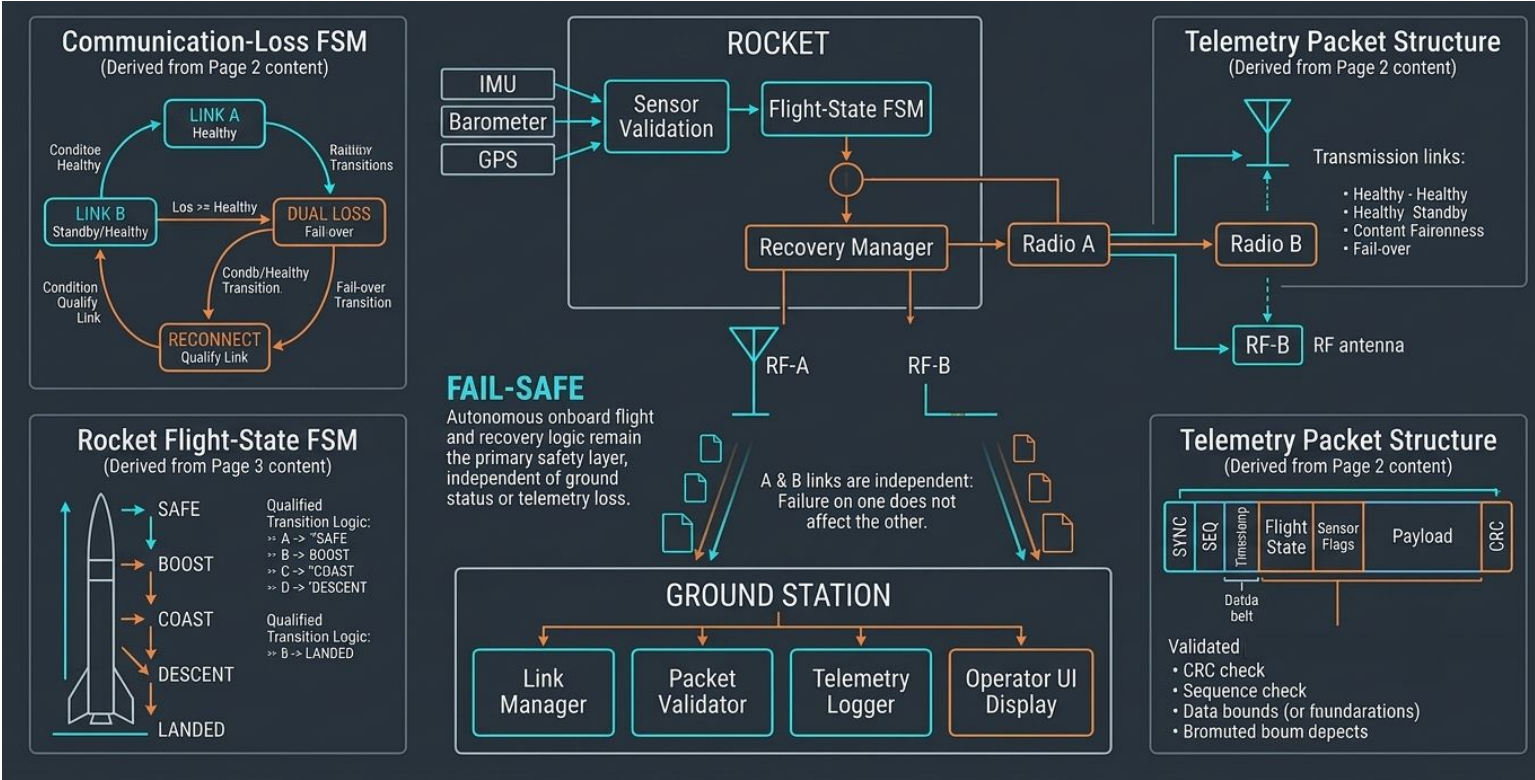
TWR (LIFTOFF / AVG)

19.4 / 8.75



GS:

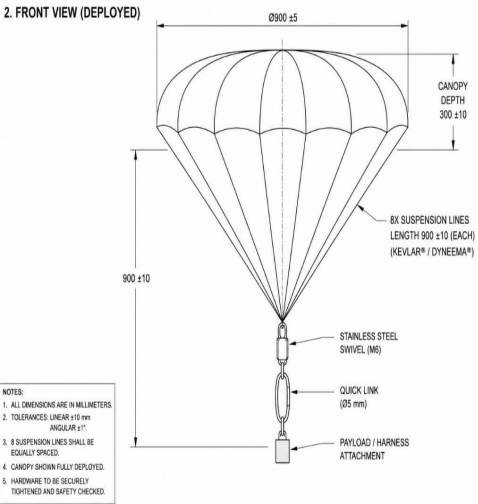
The infographic details a redundant dual-radio telemetry architecture connecting onboard rocket avionics directly to a multi-stage ground station receiver.



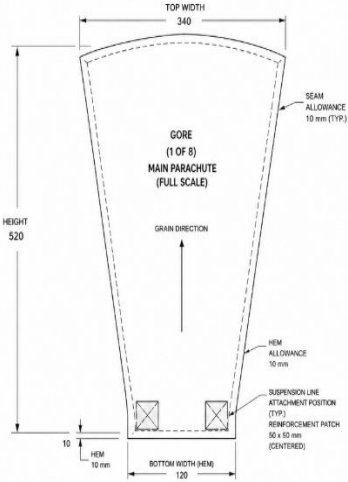
Recovery system Design



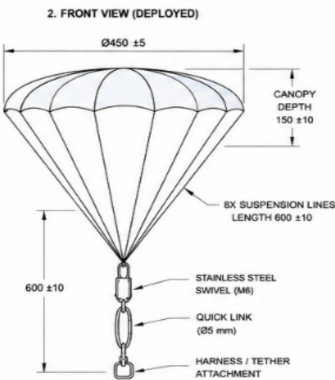
Main parachute



Gore Cutting



Droguechute



SIDE VIEW

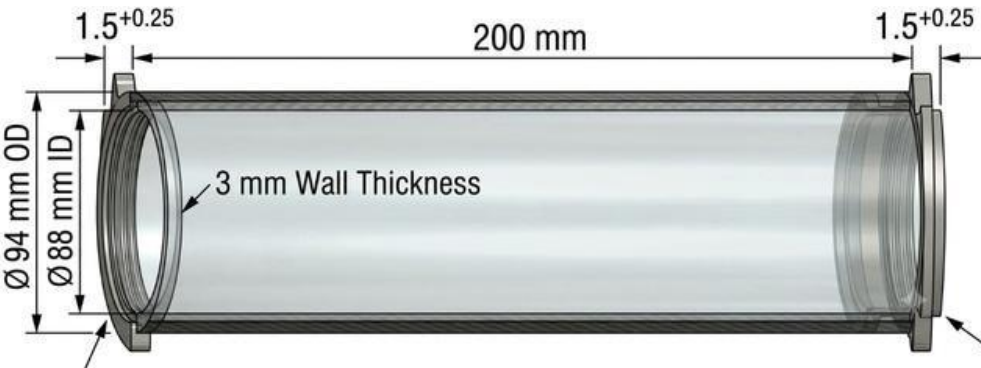
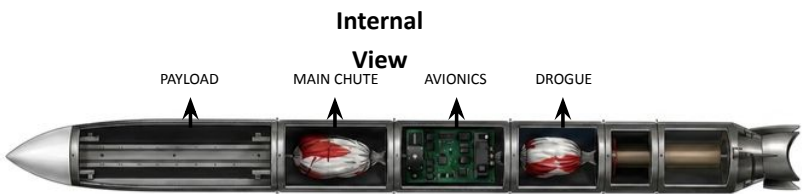


Servo

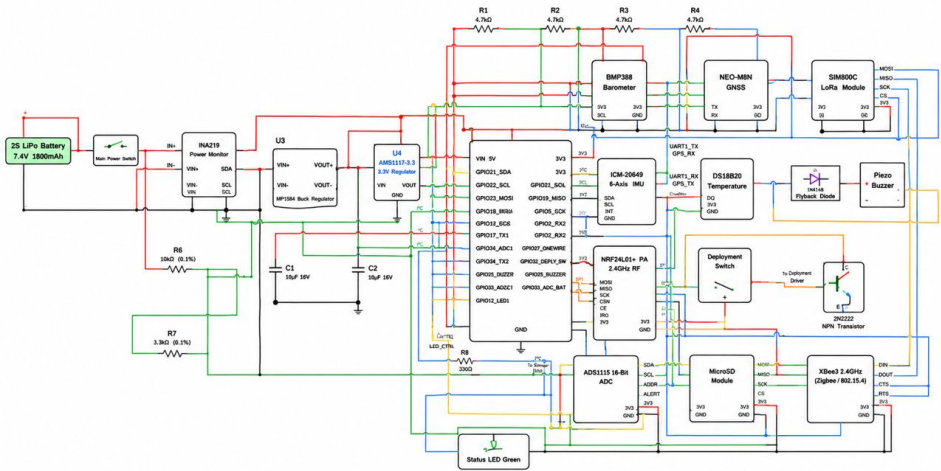


Payload System

Rocket configuration overview



Sensors:



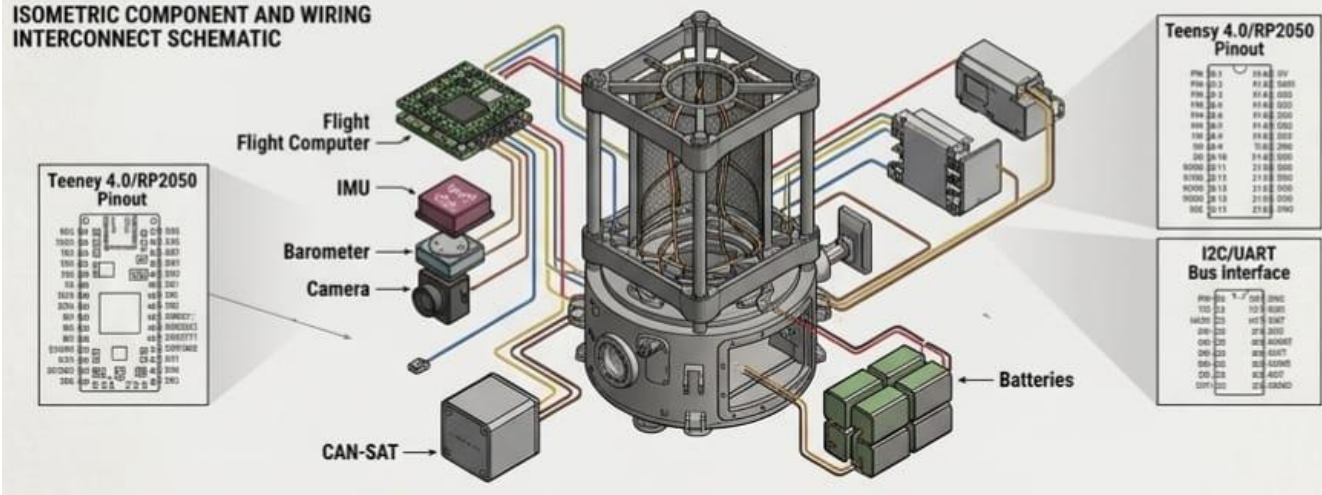
Sensor	Model	Current (mA)	Mass (g)
Flight Computer	ESP32 DevKit V1	80.00	10.0
Barometric Altimeter	BMP388	3.40	1.5
9-DOF IMU	ICM-20948	2.50	1.5
High-G Accelerometer	H3LIS331DL	1.00	1.5
GNSS Module	NEO-M10S	18.00	12.0
Temperature Sensor	DS18B20	1.50	3.0
Voltage Monitor	INA219	1.00	1.5Module (Optional)
ADC	ADS1115	0.15	1.5
Deployment Switch	Omron D2F-01F	0.00	1.6
Recovery Buzzer	Passive Piezo	0.00 (idle)	5.0
SD Card Module	MicroSD SPI	0.00 (idle)	3.0
Voltage Regulator	MP1584 Buck	5.00	3.0
Pull-up Resistors	Passive	0.50	0.5
Camera Module(Optional)	ESP32-CAM (OV2640)	120-180	8.0g
Subtotal (sensors + MCU)		113.05 mA (+120 mA)	46.1 g + 8.0g

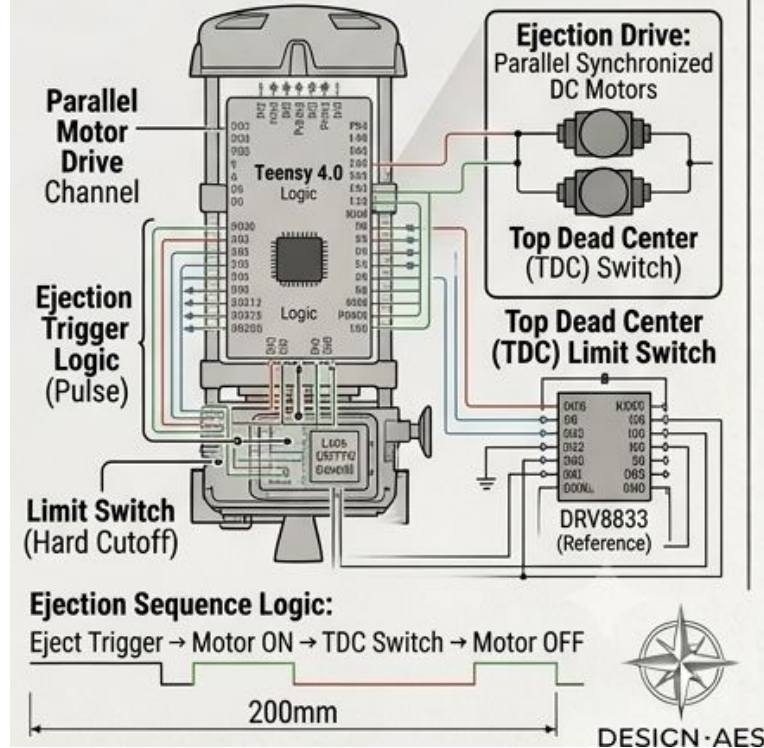
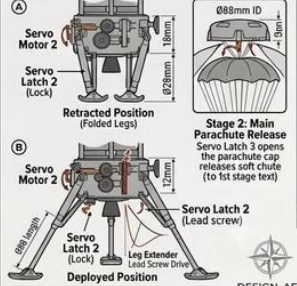
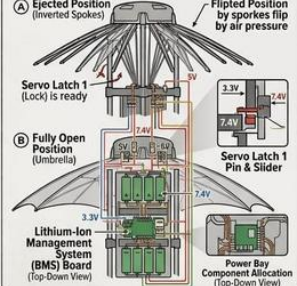
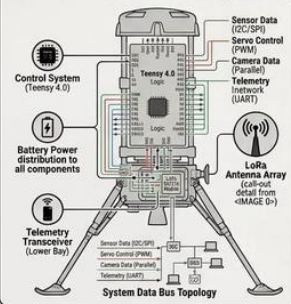
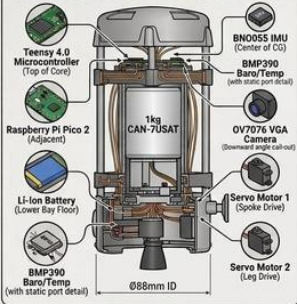
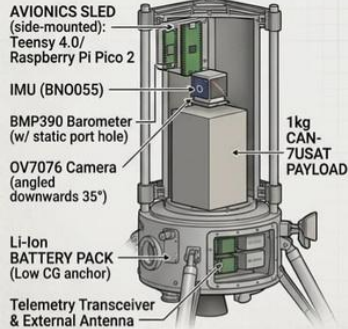
Payload Structural Configuration

MISSION TIMELINE & TELEMETRY

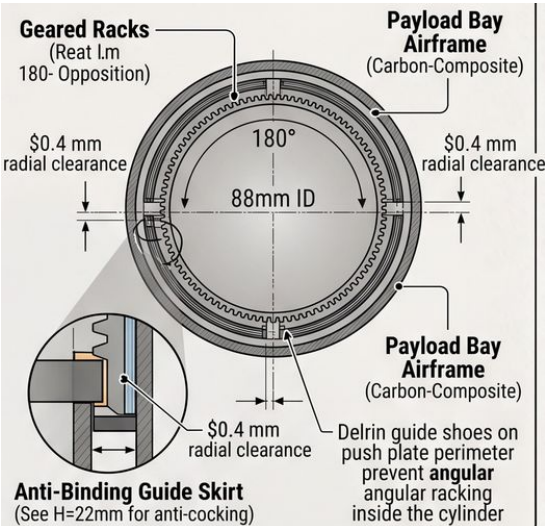
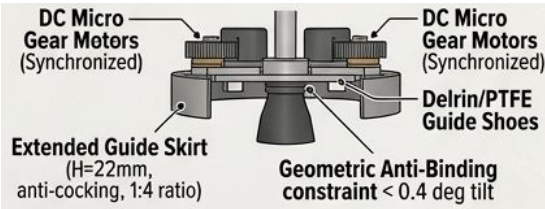


ISOMETRIC COMPONENT AND WIRING INTERCONNECT SCHEMATIC





Payload Structural Configuration



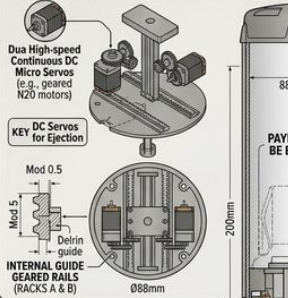
Push-Plate Ejection Integration Overview

Parameter	Specification	Function
Travel Stroke	~180–190 mm	Clears payload base out of the 200 mm tube
Drive Interface	Dual Pinion-to-Rack (Mod 0.5)	Positive, slip-free linear displacement
Ejection Speed	0.25–0.4 m/s	Clean, controlled exit without sudden impulse shock
Clearance Envelope	Ø87.2 mm disc	0.4 mm radial clearance inside Ø88 mm tube
Stroke Limit Cutoff	Microswitch / Hall sensor at top	Cuts motor power once payload clears the bay rim

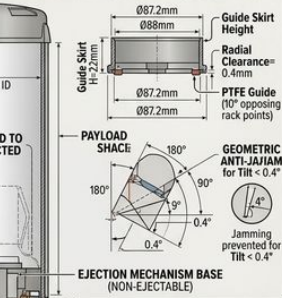
Payload Bay Structural Configuration

PAYLOAD BAY EJECTION MECHANISM (L=200mm, Ø88mm ID)

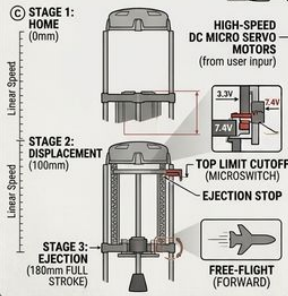
DETAIL SCHEMATIC A: CARRIAGE MECHANISM & DUAL-MOTOR DRIVE



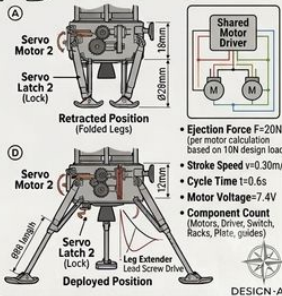
DETAIL SCHEMATIC B: ANTI-BINDING & SKIRT GEOMETRY



CAD: EJECTION STROKE TIMELINE & LIMIT CUTOFF



CAD: ELECTRICAL DRIVE & COMPONENT SPECIFICATIONS



Mass & Dynamic Load (m):

- CAN-7USAT core: 1.0 kg
- Lander frame, legs, parachute assemblies, avionics, battery: ≈ 0.35 kg
- Total mass to displace (m_{total}): ≈ 1.35 kg
- Factoring in guide friction ($\mu \approx 0.15$) and safety factor ($SF = 1.8$), design lift force:

$$F_{\text{total}} \approx 20 \text{ N} \Rightarrow F_{\text{motor}} = 10 \text{ N per motor}$$

Rack & Pinion Sizing:

- Module (m): 0.5 mm
- Pinion Tooth Count (Z): 20 teeth
- Pitch Circle Diameter (D_p): $m \times Z = 10.0$ mm (Pitch Radius $r_p = 5.0$ mm)
- Ejection Stroke (L): 180 mm (out of 200 mm bay length)

Kinematics & Motor Requirements:

- Target Separation Duration (t): ≈ 0.6 s
- Linear Separation Speed (v):

$$v = \frac{L}{t} = \frac{0.18 \text{ m}}{0.6 \text{ s}} = 0.30 \text{ m/s}$$

- Pinion Angular Velocity (ω / N):

$$\omega = \frac{v}{r_p} = \frac{0.30 \text{ m/s}}{0.005 \text{ m}} = 60 \text{ rad/s} \approx 573 \text{ RPM}$$

- Torque Required per Motor (T):

$$T = F_{\text{motor}} \times r_p = 10 \text{ N} \times 0.005 \text{ m} = 0.050 \text{ N} \cdot \text{m} \approx 0.51 \text{ kg} \cdot \text{cm}$$

- Mechanical Power Output per Motor (P):

$$P = T \times \omega = 0.050 \text{ N} \cdot \text{m} \times 60 \text{ rad/s} = 3.0 \text{ W}$$

- Motor Recommendation: Dual high-speed metal-gear micro DC gearmotors or 360° continuous micro servos rated for ≥ 0.8 kg · cm stall torque at ~ 600 RPM (running at 6.0V – 7.4V).

DETAILED CAD/SCHEMATIC: PAYLOAD BAY EJECTION MECHANISM
- MOTOR-DRIVEN PUSH PLATE SYSTEM (Ø88mm ID x 200mm)

